

SQL

PROJECT - E-Commerce

```
FORMAT(COLUMN_NAME,  
       'yyyy-mm')
```

1. Sales Performance Analysis

Write a query to analyze the monthly sales performance for the last 12 months. Your query should return the following columns:

- Month (YYYY-MM format)
- Total Sales Amount (from `SalesOrderHeader`)
- Total Number of Orders
- Average Order Value

Tables:

- `Sales.SalesOrderHeader`

Columns:

- `SalesOrderHeader.OrderDate`
- `SalesOrderHeader.TotalDue`
- `SalesOrderHeader.SalesOrderID`

2. Customer Retention

Question:

Identify the customers who have placed more than 3 orders in the last 6 months. Return the following columns:

- Customer ID
- Customer Name
- Number of Orders Placed

Tables:

- `Sales.Customer` , `Sales.SalesOrderHeader`

Columns:

- `Customer.CustomerID` , `Customer.FirstName` , `Customer.LastName` , `SalesOrderHeader.CustomerID` , `SalesOrderHeader.SalesOrderID` , `SalesOrderHeader.OrderDate`

3. Product Performance by Category

Generate a report of the top 5 best-selling products (by quantity) in each product category.
The report should include:

- Product Name
- Product Category
- Total Quantity Sold
- Total Sales Revenue

Tables:

- `Production.Product` , `Production.ProductCategory` ,
`Sales.SalesOrderDetail` , `Sales.SalesOrderHeader`

Columns:

- `Product.ProductID` , `Product.Name` , `ProductCategory.Name` ,
`SalesOrderDetail.OrderQty` , `SalesOrderDetail.LineTotal` ,
`SalesOrderHeader.OrderDate`

4. Inventory Stock Alert

Write a query to identify products that are running low on stock. List the products where the quantity in stock is less than the reorder level. The output should include:

- Product Name
- Product ID
- Quantity in Stock

Tables:

- `Production.Product`
- `Production.ProductInventory`

Columns:

- `Product.Name` , `Product.ProductID` ,
`ProductInventory.Quantity` , `Product.ReorderPoint`

5. Profit Margin Calculation

Calculate the profit margin for each product sold, by subtracting the cost of the product from its sales price. Return the following columns:

- Product Name
- Product ID
- Sales Price
- Product Cost
- Profit Margin (Sales Price - Cost)

Tables:

- `Production.Product` , `Production.ProductCostHistory` ,
`Sales.SalesOrderDetail`

Columns:

- `Product.Name` , `Product.ProductID` ,
`ProductCostHistory.StandardCost` , `SalesOrderDetail.UnitPrice`

6. Return Rate Analysis

Question:

Create a query that calculates the return rate of each product sold over the last 6 months. The return rate should be defined as the number of returned items divided by the number of sold items. Return the following columns:

- Product Name
- Product ID
- Number of Items Sold
- Number of Items Returned
- Return Rate (as a percentage)

Tables:

- `Sales.SalesOrderDetail` , `Sales.SalesOrderHeader` , `Production.Product`

Columns:

- `SalesOrderDetail.ProductID` , `SalesOrderDetail.OrderQty` ,
`SalesOrderHeader.Status` , `SalesOrderHeader.OrderDate` , `Product.Name`

7. Order Shipping Analysis

Question:

Write a query to find the average shipping time for orders in the last 6 months. The shipping time should be calculated as the difference between the order date and the ship date. Your output should include:

- Average shipping time in days
- Total number of orders shipped
- Maximum and Minimum shipping times

Tables:

- `Sales.SalesOrderHeader`

Columns:

- `SalesOrderHeader.OrderDate` , `SalesOrderHeader.ShipDate` ,
`SalesOrderHeader.SalesOrderID`

8. Cart Abandonment Rate

Write a query to calculate the cart abandonment rate. Identify the percentage of users who added items to their cart but did not complete the purchase. Consider the following:

- Abandoned Cart: Users who have items in `ShoppingCartItem` but no corresponding completed order in `SalesOrderHeader` within 48 hours.

Output: User ID , Total Items Added to Cart , Items Purchased , Abandonment Rate (percentage of items abandoned)

Tables:

- `Sales.ShoppingCartItem` , `Sales.SalesOrderHeader` ,
`Sales.SalesOrderDetail`

Columns:

- `ShoppingCartItem.CustomerID` , `ShoppingCartItem.ProductID` ,
`SalesOrderHeader.CustomerID` , `SalesOrderHeader.Status` ,
`SalesOrderDetail.ProductID`

